

18 A student immersed a plant tissue in solution S.

For the next 30 minutes, the student used a microscope to observe the cells in the plant tissue.

During this time, the cytoplasm of the cells shrunk and no longer pushed up against the cell wall. In many cells, a space appeared between the cytoplasm and the cell wall.

Which statements are correct conclusions from these observations?

- 1 The water potential in solution S was more negative than the water potential in the cytoplasm of the cells at the start of the experiment.
- 2 Without a cell wall to provide support, it is likely that many of the cells would have burst.
- 3 The clear space between the cell wall and the cytoplasm of many of the cells was filled with air.

A 1 and 3

B 1 only

C 2 and 3

D 2 only